

# Etsy

By Armani Chalmers

## ETSY

- ▶ E-commerce platform specializing in unique, handmade, and vintage items, ranging from jewelry and home décor to personalized gifts and craft supplies.



# Client Overview



MISSION  
STATEMENT



PRODUCTS



SERVICES

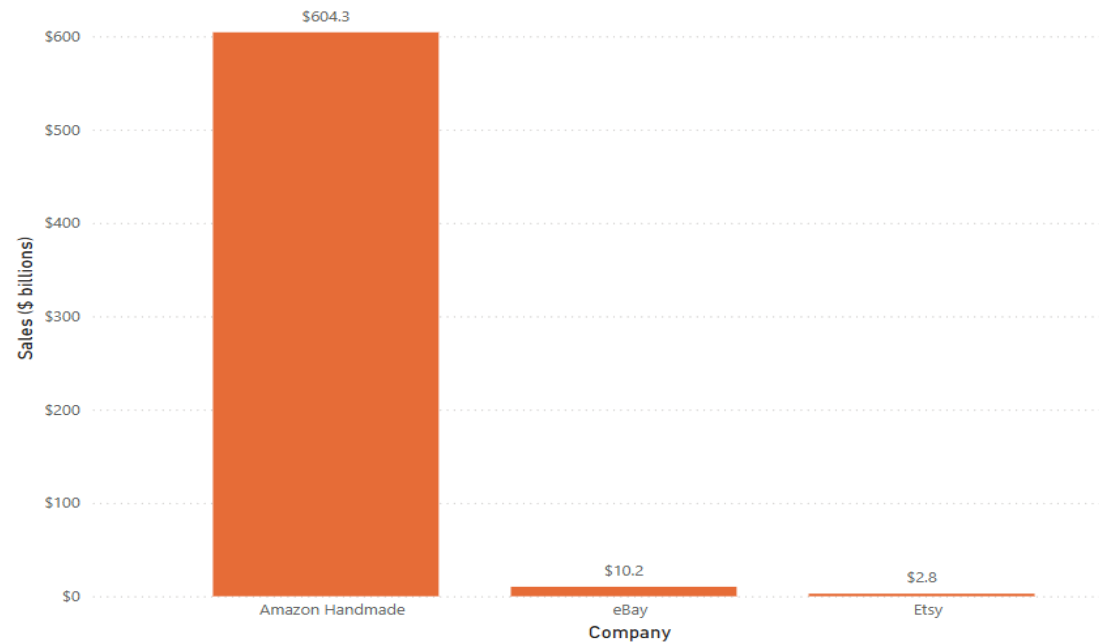


MARKETS



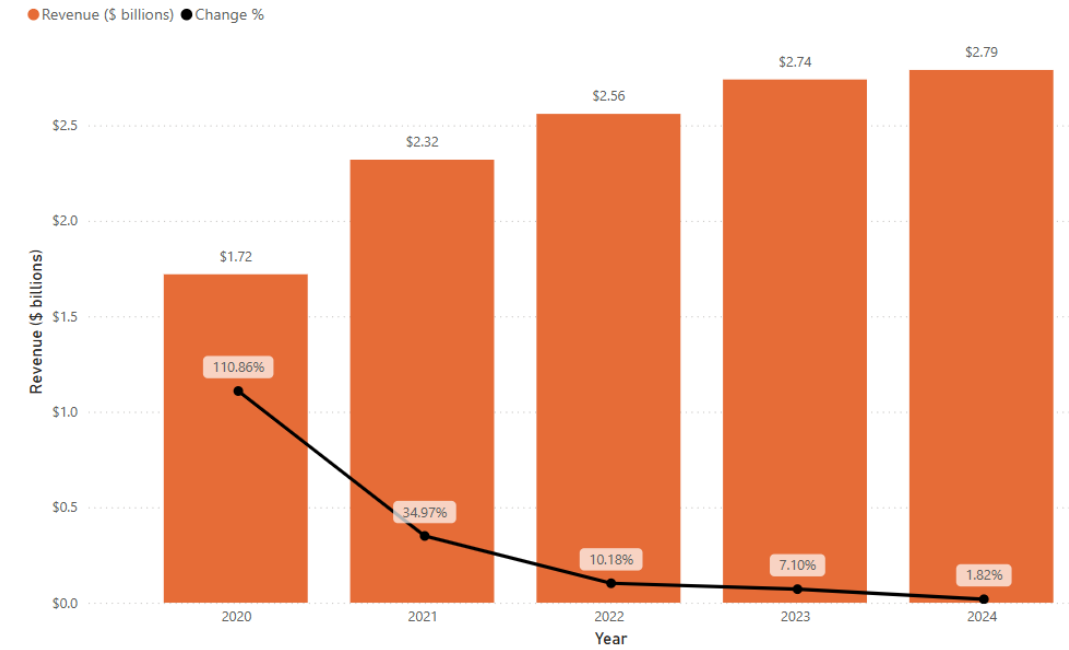
## Competitor Sales

Competitor Sales (\$ billions) by Company



## Etsy's 5-Year Revenue

Etsy's Revenue (\$ billions) and Change % by Year



# PESTEL Analysis

P

## Political

- Changes in advertising regulations and privacy laws impact how Etsy Ads can target and communicate with consumers.
- Shifts in international trade policies and taxation frameworks affect cross-border sales and seller profitability.
- Political instability in key markets can disrupt seller operations and consumer demand.

E

## Economic

- Inflation and rising living costs can reduce disposable income, leading consumers to cut back on non-essential purchases like handmade and vintage goods.
- Exchange rate volatility impacts the affordability of international sales, affecting both sellers and buyers across borders.
- Slower economic growth or recessions can tighten seller budgets for advertising and reduce overall platform engagement.

S

## Social

- Growing demand for sustainable, ethically made products aligns with Etsy's brand focus on handmade and eco-conscious goods.
- Demographic shifts, including the rise of younger, mobile-first consumers, increase the importance of personalized and mobile-optimized shopping experiences.
- Cultural trends favoring support for small businesses and local artisans create opportunities for Etsy to deepen customer loyalty.



# PESTEL Analysis

T

## Technological

- AI and machine learning are driving smarter ad targeting, personalization, and campaign optimization across the e-commerce industry.
- Increased competition from platforms investing heavily in innovation, such as Amazon and Shopify, raises the pressure on Etsy to continuously improve its ad technologies.
- Mobile commerce growth demands seamless app experiences and real-time responsiveness for shoppers and advertisers alike.

E

## Environmental

- Growing consumer demand for sustainable products creates opportunities for Etsy to highlight eco-friendly sellers and initiatives.
- Climate change awareness is driving preferences for lower-impact production, packaging, and shipping practices.
- New environmental regulations and standards could affect how products are sourced, marketed, and delivered across global markets.

L

## Legal

- Intellectual property rights enforcement is essential to prevent counterfeit goods and protect the originality of handmade and vintage items.
- Health and safety regulations, especially for products like cosmetics, jewelry, and children's goods, require strict compliance to avoid legal risks.
- Changes in international labor laws and product liability standards could impact seller requirements and cross-border sales practices.



# SWOT Analysis



**STRENGTHS**



**WEAKNESSES**



# SWOT Analysis



**OPPORTUNITIES**



**THREATS**

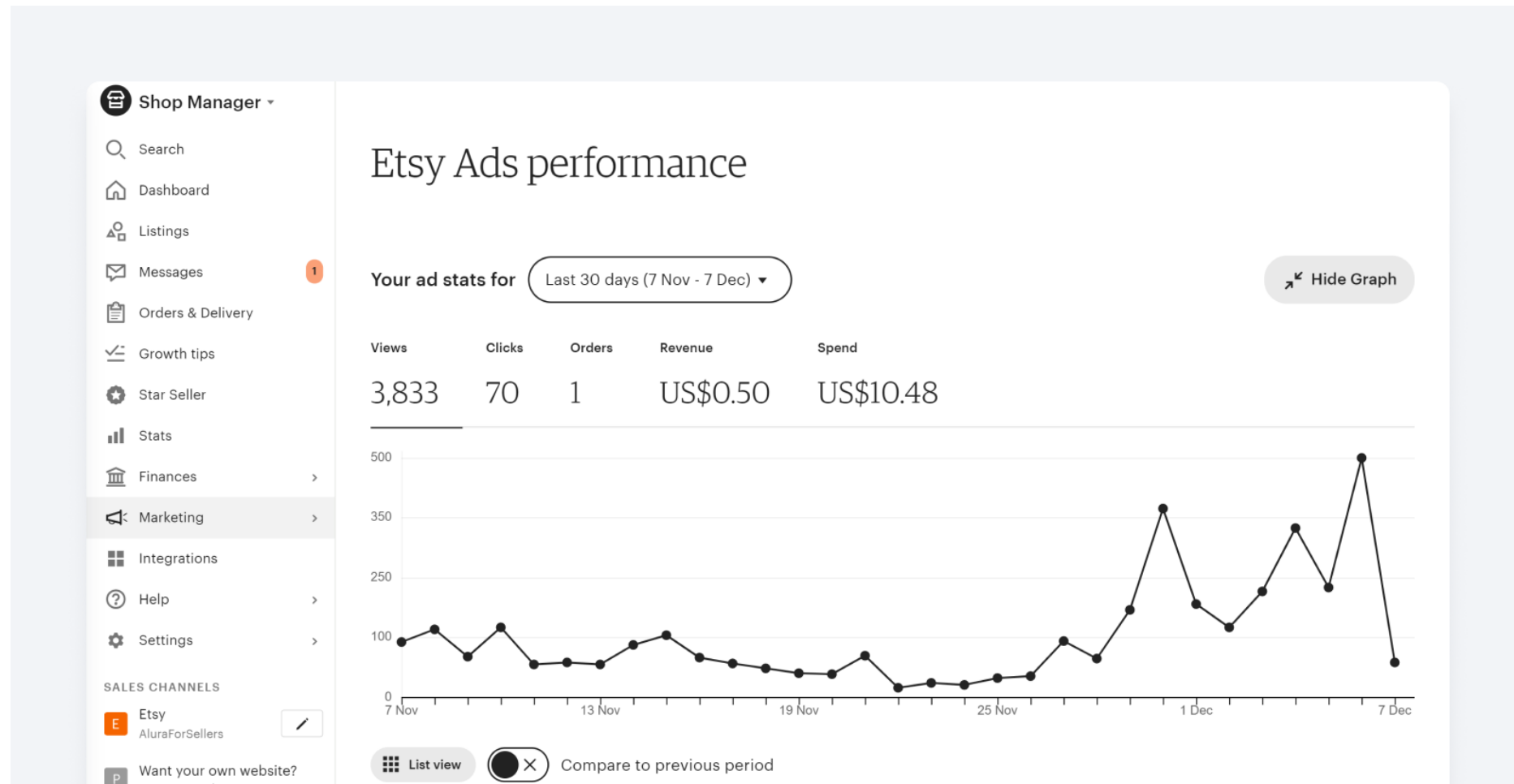




(Guitarfoto, 2020)

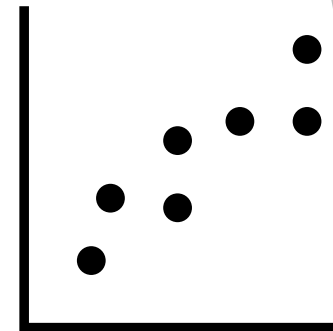
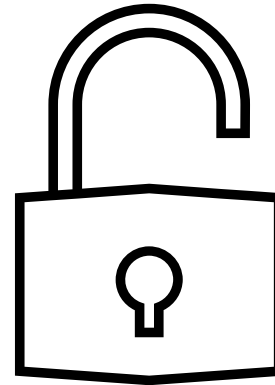
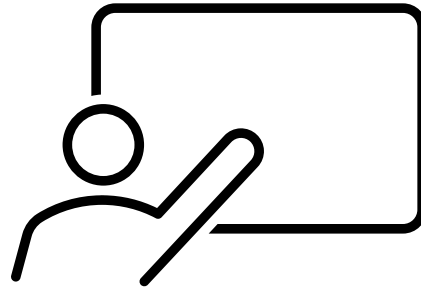
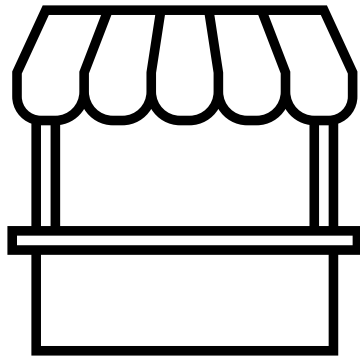
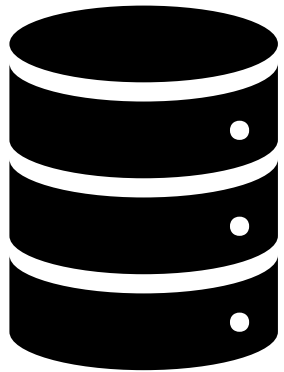


## Proposed BI Solution for Etsy Ads



(Alura, n.d.)

# Implementation Process Overview



| Phase                                                    | Timeline              | Key Milestones                                                                                             |
|----------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------|
| 1. Set Up the BI Platform and Data Warehouse             | Months 1–2            | Finalize vendor contracts, configure cloud infrastructure, and build centralized data storage.             |
| 2. Connect Data Sources Through ETL Integration          | Months 3–4            | Map, clean, and integrate ad performance, seller, and customer data using ETL pipelines.                   |
| 3. Develop Seller-Facing Dashboards and Predictive Tools | Months 5–6            | Design dashboards, set up predictive analytics, and integrate A/B testing features.                        |
| 4. Train Internal Teams and Sellers                      | Months 7–8            | Conduct training for Etsy teams; create onboarding guides, tutorials, and support resources for sellers.   |
| 5. Establish Security and Governance Protocols           | Months 7–8 (parallel) | Implement data protection, compliance standards, and transparency measures alongside training.             |
| 6. Launch, Monitor, and Iterate                          | Months 9–10           | Run pilot launch, gather feedback, roll out in phases, monitor adoption, and refine based on seller input. |

# High Level Timeline



| Area of Focus                           | Current State                                                                  | Future State                                                     | Gap                                                                 | BI Recommendation                                                                                                     |
|-----------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Real-Time Data Availability</b>      | Delayed reporting, often updated hourly or daily                               | Live, real-time dashboards for instant performance tracking      | Data latency limits timely decision-making                          | Implement real-time data streaming and live dashboard capabilities using tools like Looker or Tableau.                |
| <b>Use of External Market Data</b>      | Minimal use of external data like seasonal trends or competitor insights       | Integrated external data to inform ad targeting and budgeting    | Missed opportunities for strategic targeting and seasonal relevance | Incorporate third-party data APIs to bring in external market trends, seasonality, and competitor insights.           |
| <b>Ad Campaign Optimization Tools</b>   | No built-in AI tools for predictive analytics or automated A/B testing         | AI-driven tools that auto-optimize campaigns and test variations | Limited scalability and inefficiency in campaign performance        | Deploy AI-powered ad optimization tools that include automated A/B testing and predictive analytics.                  |
| <b>Seller Dashboard Functionality</b>   | Limited actionable insights and high manual effort required                    | User-friendly dashboard with dynamic, actionable insights        | Sellers are forced to rely on guesswork, decreasing satisfaction    | Redesign the seller dashboard to surface key metrics and offer real-time suggestions using natural language insights. |
| <b>Advertising Performance Feedback</b> | Sellers receive lagging indicators, making it difficult to adjust in real-time | Immediate feedback with suggestions for campaign adjustments     | Lack of responsiveness reduces return on ad spend (ROAS)            | Set up real-time campaign monitoring alerts and automated performance insights to help sellers adjust quickly.        |

## Gap Analysis: Required Capabilities vs. Existing Capabilities



# Gap Analysis: Processes, Tools, Staffing Required



## Processes

Data integration  
Campaign monitoring  
Seller support workflows  
Continuous BI system optimization



## Tools

Power BI  
ETL Tools  
Snowflake or BigQuery for data warehousing  
Azure or Google Cloud for hosting

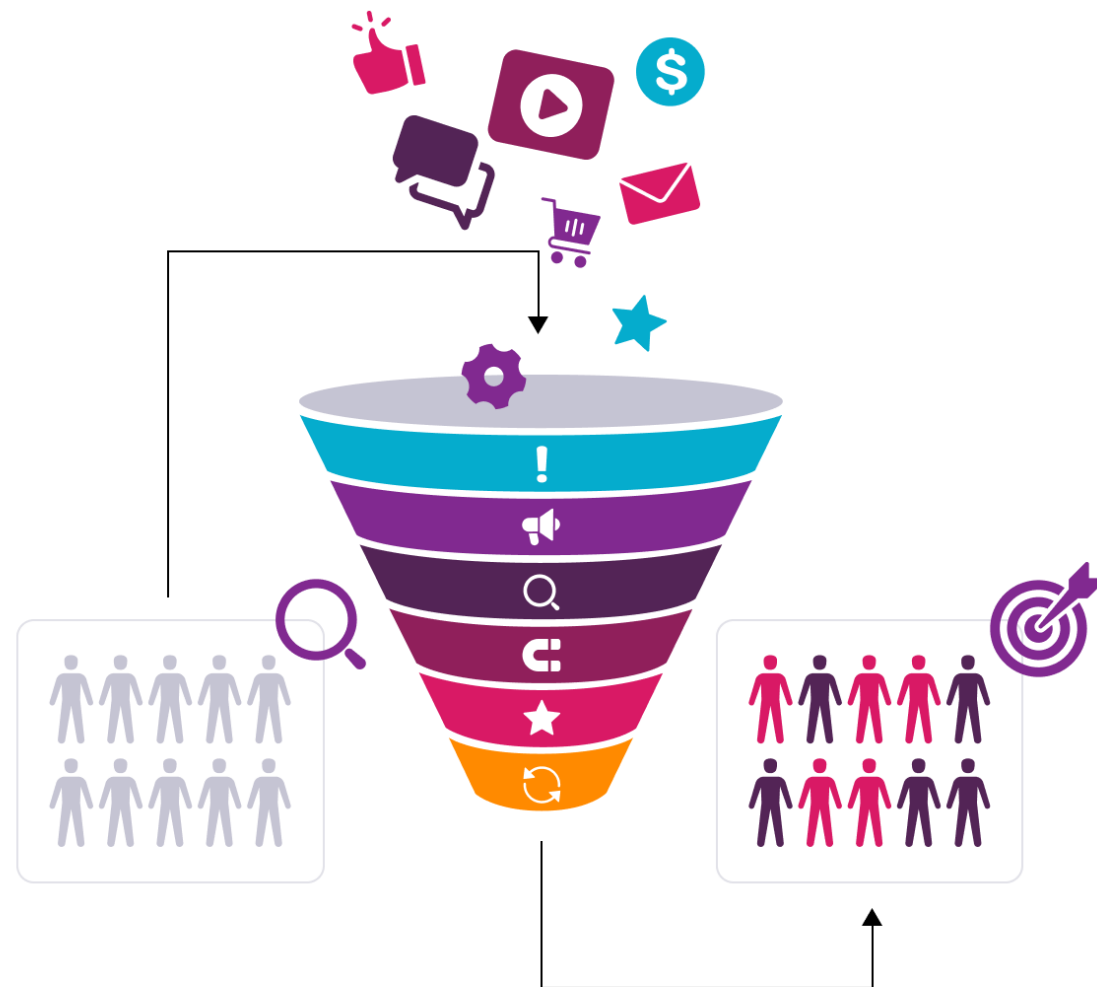


## Staffing

BI analysts  
Data engineers  
Cloud architects  
Seller support trainers  
Compliance and QA leads



## Data Collection Process



## Data Management Infrastructure



# METHODS OF DATA MINING



**Classification**



**Clustering**



**Regression Analysis**



**Association Rule Mining**





**Descriptive**  
Explains what  
happened.



**Diagnostic**  
Explains why it  
happened.



**Predictive**  
Forecasts what  
might happen.



**Prescriptive**  
Recommends an  
action based on  
the forecast.

# Data Analysis Techniques





# Etsy Ads performance

Your ad stats for

Last 30 days (Apr 17 - May 17) ▼

Views

Clicks

Orders

Revenue

Spend

93.6K

1,129

14

\$947.70

\$310

## Monitoring and Reporting

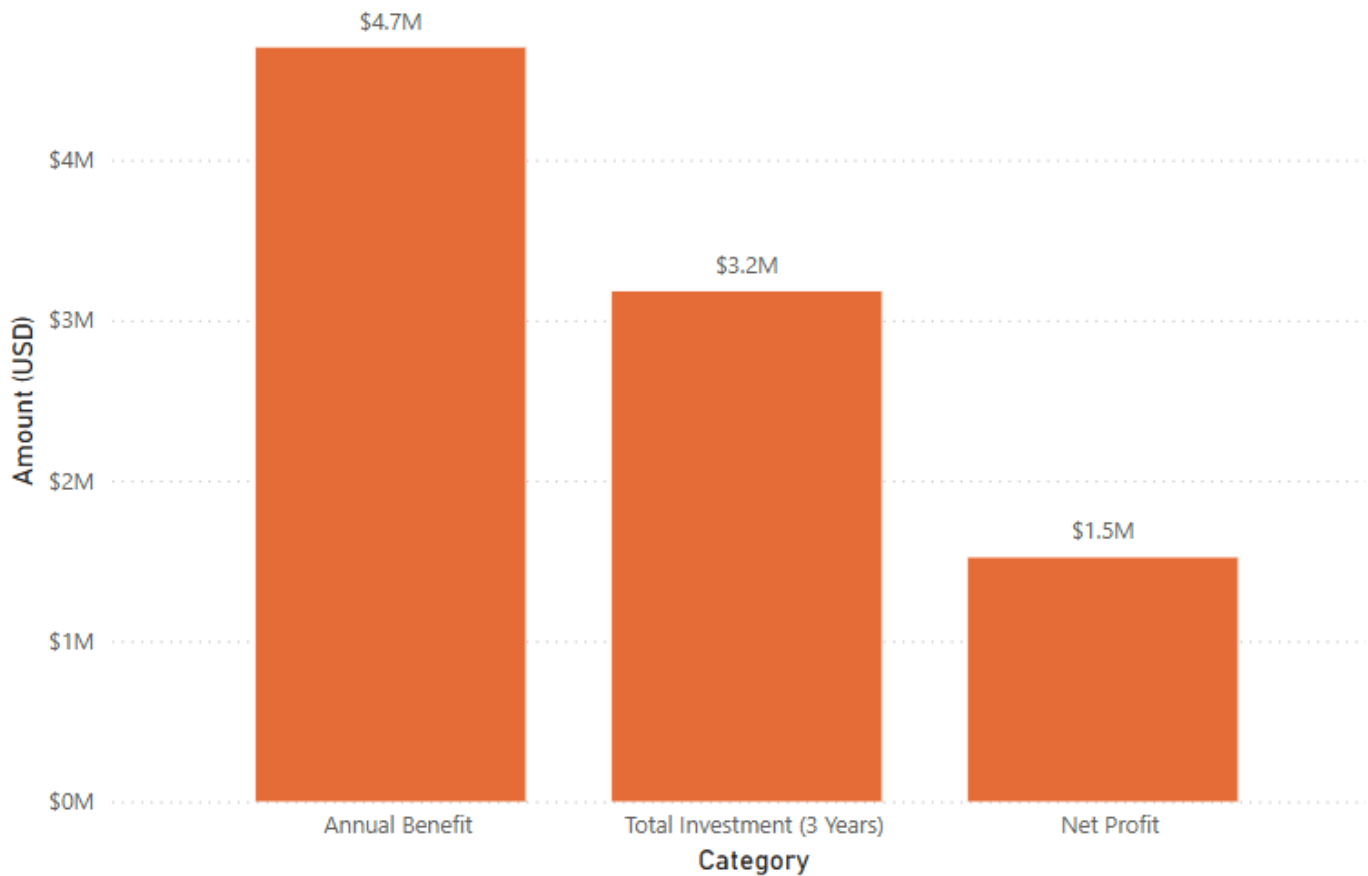


| Category                        | Requirement                                      | Quantity              | Estimated Cost |
|---------------------------------|--------------------------------------------------|-----------------------|----------------|
| Data Collection                 | Real-time streaming tools (Kafka, etc.)          | 1 system setup        | \$140,000      |
|                                 | External data API integrations                   | 3-5 APIs              | \$50,000       |
|                                 | Data Engineers                                   | 2 FTEs                | \$240,000/year |
| Data Management                 | Cloud Data Warehouse (Snowflake, BigQuery)       | 1 subscription        | \$280,000/year |
|                                 | Data Governance Tools                            | 1 license             | \$90,000/year  |
|                                 | Cloud Infrastructure (AWS, GCP)                  | Usage-based           | \$160,000/year |
| Data Mining/AI                  | Predictive Analytics Platform (Databricks, etc.) | 1 license             | \$230,000/year |
|                                 | Data Scientists                                  | 2 FTEs                | \$280,000/year |
|                                 | ML Engineer                                      | 1 FTE                 | \$140,000/year |
| Analytics & Reporting           | BI Tool License (Looker, Tableau)                | 1 license             | \$200,000/year |
|                                 | Seller Dashboard Customization                   | Initial build         | \$70,000       |
|                                 | Seller and Staff Training Programs               | 4 workshops           | \$40,000       |
| Monitoring                      | Automated Reporting & Alerts                     | 1 setup               | \$50,000       |
|                                 | KPI Standardization Project                      | 1 internal initiative | \$30,000       |
|                                 | Support & Monitoring Staff                       | 1-2 FTEs              | \$160,000/year |
| Compliance & Security           | Data Privacy Compliance (GDPR, CCPA) Audits      | 2 audits              | \$40,000       |
|                                 | Security Tools and Encryption                    | 1 implementation      | \$100,000      |
| Change Management               | Change Management Lead                           | 1 FTE                 | \$120,000/year |
|                                 | Communication & Rollout Campaign                 | Organization-wide     | \$50,000       |
| Total Estimated First-Year Cost |                                                  |                       | ~\$3.1 million |

# Type, Quantity, and Cost Requirements



Projected 3-Year ROI



Investment Breakdown

| Cost Area                              | 3-Year Total | Details                                                                    |
|----------------------------------------|--------------|----------------------------------------------------------------------------|
| BI Software Licenses                   | \$525,000    | Enterprise-level tools such as Power BI or Tableau                         |
| Cloud Infrastructure                   | \$660,000    | Hosting for data processing, storage, and scalability                      |
| Data Warehouse Setup                   | \$540,000    | Centralized system for storing ad and performance data                     |
| ETL Tools and Integration              | \$360,000    | Connecting Etsy's ad platform with various data sources                    |
| Dashboard Development and AI Tools     | \$750,000    | Creating seller-facing dashboards and predictive models                    |
| Training and Seller Onboarding         | \$300,000    | Tutorials, workshops, and support materials                                |
| Security and Data Governance           | \$300,000    | Ensuring compliance with privacy laws and data protection protocols        |
| Maintenance and Upgrades               | \$270,000    | Regular updates, performance monitoring, and technical support             |
| Consulting and Implementation Services | \$225,000    | BI experts and analytics consultants to guide deployment and customization |
| Total Investment                       | \$3,180,000  |                                                                            |

Projected Benefits

| Benefit                      | Estimated Annual Value | Explanation                                                                  |
|------------------------------|------------------------|------------------------------------------------------------------------------|
| Increased Seller Retention   | \$1,000,000            | Enhanced trust and transparency encourage continued use of Etsy Ads          |
| Improved Ad Conversion Rates | \$1,200,000            | Predictive tools lead to more effective targeting and higher sales per click |
| Growth in Ad Revenue         | \$1,500,000            | Satisfied sellers are likely to invest more in advertising                   |
| Operational Cost Savings     | \$400,000              | Reduction in support requests and manual troubleshooting                     |
| Decreased Seller Churn       | \$600,000              | Retention of top-performing sellers reduces revenue loss                     |
| Total Annual Benefit         | \$4,700,000            |                                                                              |

ROI Calculation

| Category                   | Amount (USD) | Description                                                     |
|----------------------------|--------------|-----------------------------------------------------------------|
| Annual Benefit             | \$4,700,000  | Total estimated annual benefits from the BI solution            |
| Total Investment (3 Years) | \$3,180,000  | Total cost of investment over three years                       |
| Net Profit                 | \$1,520,000  | \$4,700,000 - \$3,180,000                                       |
| ROI (Annual)               | 47.80%       | $(\$1,520,000 \div \$3,180,000) \times 100$                     |
| Raw 3-Year ROI             | 143.40%      | $47.8\% \times 3$ (without adjusting for time value or ramp-up) |
| Adjusted 3-Year ROI        | ~84%         | More realistic estimate accounting for phase-in and ramp-up     |

## Conclusion

Etsy

A black line graph is overlaid on the word 'Etsy'. The graph features a vertical y-axis and a horizontal x-axis. A jagged line starts at the origin, trends upwards with minor fluctuations, and ends in an arrowhead pointing towards the top right. The word 'Etsy' is written in a large, orange, serif font, with the graph's axes and line passing through the letters 't' and 's'.